

Strategic Analysis of Embedded Small Language Model Inference on Solid State Drive Controllers

Executive Summary

The transition of the solid-state drive from a passive repository of binary data to an active computational endpoint represents a structural shift in the data center and edge computing hierarchy. As large language models scale toward trillions of parameters, the traditional memory-processor interface is encountering a physical and economic boundary known as the memory wall, where High Bandwidth Memory (HBM) capacity cannot keep pace with the data-intensive nature of inference, particularly in Retrieval-Augmented Generation (RAG) and long-context reasoning. This report investigates the feasibility of embedding Small Language Models (TinyLMs) directly within KIOXIA solid-state drive (SSD) controllers, leveraging the high density of 10th-generation BiCS FLASH™ and the low latency of XL-FLASH™ Storage Class Memory (SCM). The analysis identifies that the current technology trajectory is moving toward "Near-Zero DRAM" architectures, where software-defined layers such as AiSAQ™ allow for billion-scale vector searches directly from flash media with minimal memory footprints. Competitive pressure from integrated computational storage players like ScaleFlux, who are adopting ARM Cortex-R82 64-bit real-time processors, necessitates a strategic shift for KIOXIA to integrate dedicated neural processing units (NPUs) or systolic arrays within its controller silicon. Technical feasibility is anchored by extreme quantization techniques, specifically 1.58-bit ternary models (BitNet), which reduce model weights to a scale compatible with the sub-100MB memory constraints of a storage controller. Market signals indicate that privacy, data sovereignty, and the 200x energy efficiency advantage of local storage compute over remote DRAM accesses are driving enterprise demand for intelligent, queryable storage solutions.

KIOXIA Technology Trajectory

KIOXIA's trajectory is governed by a focus on maximizing bit density while simultaneously engineering the latency out of the memory hierarchy. The evolution from early NAND flash to the current BiCS FLASH™ 3D architecture has provided the foundational capacity required for modern AI workloads, but the company's recent focus on Software-Enabled Flash (SEF) and AiSAQ™ signifies a pivot toward intelligent data management.

Evolution of BiCS FLASH™ and CMOS Directly Bonded to Array

The cornerstone of KIOXIA's trajectory is the BiCS FLASH™ 3D technology, which has moved through ten generations of vertical and lateral scaling. The 8th generation marked a significant departure from traditional manufacturing by introducing CMOS directly Bonded to Array (CBA) technology. This process involves the separate fabrication of the CMOS logic wafer and the memory cell array wafer, which are then bonded together. This allows for the logic layer to be manufactured using more advanced process nodes without being subjected to the thermal

cycles required for NAND cell formation, enabling higher interface speeds and more complex logic functions within the drive.

The roadmap for 2026-2027 targets the mass production of 10th-generation BiCS FLASH™, which is projected to achieve 332 layers. This represent a 1.5x increase in layer count over the 8th generation, which is essential for storing the massive vector databases associated with generative AI. The integration of CBA technology facilitates interface speeds up to 3.6 Gbps, providing the necessary bandwidth to support on-device AI accelerators.

Generation	Stacking Architecture	Key Innovations	Targeted Application
BiCS 8	~218 Layers	CBA, On Pitch Select (OPS)	Enterprise SSDs, 2Tb QLC
BiCS 9	Developing	Enhanced Bit Density, Power Efficiency	Data-intensive AI, High-speed interfaces
BiCS 10	332 Layers	Industry-leading layer count	Generative AI training/inference infrastructure

XL-FLASH™ and the Latency-Critical Tier

While standard NAND provides capacity, XL-FLASH™ serves as KIOXIA's high-performance tier, bridging the gap between DRAM and traditional flash. The architecture uses a unique 16-plane design that addresses word and bit lines in short dimensions, reducing read latency to less than 5 microseconds. The second generation of XL-FLASH™, supporting MLC (Multi-Level Cell) functions, is currently in mass production, providing a balance of speed and cost for applications such as high-frequency trading and AI vector search.

KIOXIA is pushing this tier further with a Super High IOPS SSD project, which demonstrated an emulator capable of delivering over 100 million IOPS. This project, in conjunction with the NVIDIA Storage-Next™ initiative, aims to allow GPUs to access flash memory directly as an expansion of HBM. By 2027, KIOXIA intends to reach the 100 million IOPS milestone using PCIe 7.0 interfaces, which would effectively treat storage as a near-memory resource for AI accelerators.

Software-Enabled Flash and Intelligent Data Management

KIOXIA's position on intelligent storage is heavily influenced by the Software-Enabled Flash (SEF) project, an open-source initiative under the Linux Foundation. SEF allows storage developers to exert direct control over latency, data placement, and workload isolation, effectively removing the "black box" nature of traditional SSD firmware. This is critical for embedding TinyLMs, as the model's inference tasks can be prioritized and scheduled alongside standard I/O operations to prevent latency spikes.

Technology	Project Type	Core Objective	Status
SEF	Open Source (Linux Foundation)	Host-controlled storage management	Production Ready
AiSAQ™	Open Source Software	SSD-optimized Approximate Nearest Neighbor Search	Integrated into Milvus 2.6.4+
RocksDB Plug-in	Software Optimization	Minimizing Write Amplification Factor (WAF)	Demonstrated on XD8 Series

The AiSAQ™ (All-in-Storage ANNS with Product Quantization) technology represents the most direct link to on-device AI inference. AiSAQ™ allows billion-scale vector searches to be performed directly from the SSD, bypassing the need to load the entire index into DRAM. KIOXIA has demonstrated scaling this to 4.8 billion vectors on a single server, with future targets in the trillion-vector range.

Relevant Patent Landscape and R&D Focus

KIOXIA's patent portfolio from 2020 to 2025 shows a distinct shift toward reliability intelligence and 3D heterogeneous integration. Analysis from PatSnap Eureka identifies KIOXIA (formerly Toshiba Memory) as a major assignee in fields such as reconfigurable SSD storage pools and workload-adaptive over-provisioning. These patents suggest that KIOXIA is developing the internal logic required for the SSD to autonomously manage its resources based on the needs of AI workloads.

Key innovation clusters in the last five years include:

- **Predictive Failover Management:** Per-block failure prediction using on-chip AI to move reliability logic into the firmware layer.
- **3D Controller Stacking:** Research into heterogeneous integration where the controller logic is stacked directly atop or beneath the memory array, reducing physical trace lengths and power consumption.
- **Content Addressable Memory (CAM) NAND:** Patents related to using NAND arrays for direct search operations, mimicking the behavior of associative memories required for AI lookups.

Partnerships and Strategic Investments

KIOXIA's primary partnership remains the joint venture with SanDisk Corporation (now part of Western Digital), which has been extended through 2034. This collaboration provides the economies of scale necessary for R&D in the Yokkaichi and Kitakami plants, which accounts for approximately 29% of the world's flash memory production capacity. Furthermore, KIOXIA's collaboration with NVIDIA on the cuVS library and Storage-Next™ initiatives signals a deep integration into the enterprise AI ecosystem, positioning KIOXIA SSDs as a fundamental tier in the NVIDIA-led AI data center architecture.

Competitive Landscape: Computational Storage

The computational storage market is undergoing a transition from niche academic curiosity to a structural component of AI infrastructure, with a projected market size of \$4.30 billion by 2032. The competitive field is split between established memory giants and agile startups focusing on specialized controllers.

Major Players and Public Progress

Samsung and ScaleFlux are currently leading the market in terms of product footprint and market share. Samsung's SmartSSD remains a pivotal product, integrating an FPGA-based processor for in-storage analytics and AI inference. However, there is a clear industry trend toward ASIC-based solutions to maximize power efficiency.

Competitor	Primary AI/Compute Direction	Recent Developments
Samsung	FPGA-based SmartSSDs and HBM dominance	Acquisition of SanDisk patents to bolster SmartSSD efficiency. Focus on HBM shortages persisting through 2027.
ScaleFlux	ASIC-based CSDs with transparent compute	Adoption of ARM Cortex-R82 for next-gen AI storage; CSD5000 series sampling.
SK Hynix	Processing-In-Memory (AiM) and HBM4	Focus on HBM market leadership to address GPU memory bottlenecks.
Micron	HBM3e and CXL Memory expansion	Leveraging CXL protocols for pooled memory; strong HBM/DRAM exposure.
ScaleFlux	Computational Storage Controllers	Demonstration of 14 GB/s throughput with integrated compression engines.

Startups and Intelligent Storage Solutions

A new wave of startups is addressing the inefficiencies of traditional block storage for AI:

- **ScaleFlux:** While expanding rapidly, they remain the primary innovator in hardware-native write reduction and transparent compression, claiming 3x performance-per-watt over traditional KIOXIA and Samsung drives.
- **Nyriad:** Specializing in GPU-accelerated storage and high-availability systems for big data workloads.
- **Eideticom:** Developers of the NoLoad® computational storage platform, which offloads intensive tasks to NVM-based accelerators.
- **Phison:** Emerging as a strong competitor in high-end controllers (e.g., SM2508), targeting low power (7W) and high sequential performance (14 GB/s) for mobile AI applications.

Existing Products for ML Inference at the Controller

The most prominent existing products in this domain include the Samsung SmartSSD and the ScaleFlux CSD series. ScaleFlux's CSD5000 series, powered by the FX5016 controller, integrates inline data compression and metadata management to reduce host CPU overhead during AI training and inference. In contrast, KIOXIA has focused on the software layer via AiSAQ™, which leverages standard enterprise SSDs (like the CM9 series) rather than a bespoke computational controller, though this software allows for AI-native search functions that mimic dedicated computational storage behavior.

Technical Feasibility: TinyLM on SSD Controllers

Embedding Small Language Models (TinyLMs) on an SSD controller requires navigating extreme constraints in terms of silicon area, power consumption, and thermal management. The feasibility is enabled by the convergence of 64-bit real-time processors and extreme model

compression.

Hardware Constraints of Modern SSD Controllers

SSD controllers are architected for deterministic performance and reliability, not general-purpose high-performance computing. They operate in a tightly regulated thermal environment, often without active cooling in client form factors.

Hardware Metric	Typical Enterprise SSD Controller Constraint	Technical Impact on TinyLM
Processor Type	32-bit/64-bit Real-time (e.g., ARM Cortex-R82)	R82 provides 64-bit addressing for >4GB RAM but requires MMU for Linux/AI.
Clock Frequency	1.0 GHz to 1.8 GHz	Significantly lower than host CPUs; limits tokens-per-second to low single digits.
On-chip SRAM	< 1 MB per core (Tightly Coupled Memory)	Insufficient for even small model weight tiles; requires constant swapping.
System DRAM	1 GB - 4 GB (Internal or Host-mapped)	Primary bottleneck for weight storage; requires aggressive quantization.
Power Budget	7 W (Mobile) to 25 W (Enterprise)	Limits sustained high-throughput neural processing.

The ARM Cortex-R82 is the first realistic candidate for hosting TinyLMs. It offers an MMU to run rich operating systems like Linux, which is necessary for hosting AI frameworks, while maintaining real-time deterministic control for storage tasks. The processor's Neon™ SIMD technology can be utilized to accelerate matrix-vector multiplications, the core operation of transformer inference.

State of the Art in Quantized Inference

To fit a language model into the sub-100MB memory footprint typical of a controller's available buffer, quantization must reach the 1-bit or 2-bit level.

- **BitNet b1.58 (Ternary Quantization):** This paradigm maps weights to values of $\{-1, 0, 1\}$. It reduces the memory required for weights by orders of magnitude and replaces expensive floating-point multiplications with simple additions and subtractions.
- **GGUF and llama.cpp:** These formats are optimized for CPU-based inference and allow for the offloading of model components between different storage tiers. Studies on Meta's Llama 3.2 3B model show a 68.66% reduction in size using 4-bit quantization, enabling mobile execution.
- **TQ1_0 and TQ2_0:** Specialized packing methods for ternary values. TQ1_0 (5 trits in 8 bits) is highly memory-efficient, while TQ2_0 is optimized for computer memory boundaries to provide faster inference.

Viable Model Architectures and Memory Footprints

For an SSD controller, the following model scales and architectures are technically viable if combined with aggressive quantization:

Model Architecture	Parameter Count	Quantization	Effective Size	Latency/Performance
Llama-3.2-1B	1 Billion	2-bit / Ternary	~200 MB - 300 MB	Acceptable for offline indexing
Phi-3.5-mini	3.8 Billion	1.58-bit (BitNet)	~600 MB	Requires 1GB+ controller DRAM
TinyML / MobileNet	< 100 Million	8-bit / 4-bit	< 50 MB	Real-time; limited reasoning

Natural language queries on a storage controller would likely target a 1B parameter model at 2-bit precision. While inference might be slow (approximately 0.5 to 1 token per second), it is sufficient for background metadata generation, document summarization, and semantic tagging of stored files.

"In-Storage Processing" in Industry and Academia

The academic and industry focus has shifted from simple computational storage toward CXL-driven memory expansion.

- **SolidAttention:** A project that offloads the Key-Value (KV) cache of LLMs to the SSD, reducing the DRAM footprint by up to 98%. This allows for extremely long context windows (128k+ tokens) on memory-constrained devices.
- **ExPAND:** A CXL-based ML prefetcher that uses machine learning on the SSD controller to predict host CPU memory access patterns, reducing latency by 9.0x to 14.7x.
- **oLLM:** A lightweight library that streams model components dynamically between GPU VRAM, system RAM, and SSD, bypassing VRAM limitations.

Market Signals and Use Cases

The demand for intelligent, queryable storage is driven by the explosive growth of unstructured data and the high cost of centralized AI processing. Market research indicates a clear shift toward tiered storage and AI-enhanced management platforms.

Enterprise Use Cases for Queryable Storage

Enterprise leaders are increasingly focused on extracting value from core business data without moving it into expensive, centralized data lakes.

- **Legal Discovery and Compliance:** Searching millions of documents for semantic intent rather than keywords. AI-powered agents can interact directly with Google Drive or local SSDs to summarize documents and draft responses.
- **Healthcare Records:** Using TinyML for real-time analysis of activity data and clinical records locally to preserve patient privacy.
- **Financial Audit Trails:** Identifying anomalies in massive transactional logs at the storage layer, reducing the burden on central servers.
- **Distributed Cyber-Resilience:** SSDs that use AI to detect ransomware patterns in I/O streams or network traffic before data is compromised.

Privacy and Sovereignty Drivers

Privacy and security are the top challenges for 44% of IT leaders surveyed. On-device storage intelligence is preferred over cloud-based indexing for several reasons:

- **Data Leakage Prevention:** Organizations do not want to send proprietary data to public LLMs just for inference.
- **Compliance:** Regulations like GDPR and HIPAA require strict controls on data movement. Local processing ensures user data remains at the source.
- **Offline Functionality:** Enabling AI intelligence in environments with limited or no internet access (e.g., edge, space, or secured facilities).

Adjacent Products and Market Trends

The "AI PC" era is driving the integration of AI functions directly into standard laptops and smartphones, which in turn increases the demand for high-capacity, high-performance flash memory.

- **AI-Powered Desktop Search:** Gemini and Copilot-like agents that need local vector databases to function efficiently.
- **STaaS (Storage-as-a-Service):** Projected to reach \$41.2 billion by 2031, with AI-enhanced management and automated tiering as key differentiators.
- **Hybrid Storage Architectures:** The cost differential between all-flash and HDD architectures sustains demand for tiered systems that use AI for intelligent data movement.

KIOXIA's Positioning and Gaps

KIOXIA is currently in a state of rapid growth driven by the AI supercycle, with stock prices rising approximately 540% in 2025 due to strong NAND demand. However, the company faces critical strategic gaps in the hardware-software integration required for on-device AI.

KIOXIA's Strengths and Vulnerabilities

Strengths:

- **BiCS FLASH™ Density:** The ability to produce 332-layer NAND provides a cost and capacity advantage for massive vector databases.
- **XL-FLASH™ Latency:** The deterministic performance of SCM is a key enabler for the "Near-GPU" cache tier, reducing the reliance on expensive HBM.
- **Open-Source Software Leadership:** Initiatives like AiSAQ™ and SEF have established KIOXIA as a software innovator in the storage stack.
- **Manufacturing Scale:** A 29% world share in production bits ensures that KIOXIA can move the market by adopting new standards.

Vulnerabilities:

- **Lack of Dedicated AI Controller IP:** While ScaleFlux and Samsung have bespoke computational controllers, KIOXIA relies on standard controllers with software offloads.
- **No HBM Portfolio:** Competitors like Samsung and SK Hynix benefit from the HBM "boom," while KIOXIA is vulnerable to the cannibalization of storage budgets by memory

spend.

- **Regional R&D Concentration:** Most R&D is localized in Japan (Yokohama, Yokkaichi) and the US, with limited research presence in emerging European hubs.

Strategic Gaps for On-Device AI

1. **Hardware-Level Tensor Acceleration:** KIOXIA must move beyond software ANNS and integrate dedicated tensor cores or systolic arrays into its controller ASIC to compete with the 3x performance-per-watt advantage of computational storage specialists.
2. **Integrated KV-Cache Management:** Developing firmware that natively manages KV-cache offloading (similar to SolidAttention) would make KIOXIA the preferred vendor for the AI PC market.
3. **End-to-End RAG Systems:** While AiSAQ™ is a strong component, KIOXIA lacks a unified hardware-software appliance that competes with the NVIDIA-ScaleFlux integrated storage targets.

Key Decision-Makers in KIOXIA R&D

Leadership in technology development is concentrated in the Frontier Technology R&D Institute and the SSD Division.

Name	Role	Location	Key Area of Influence
Hideshi Miyajima	Managing Executive Officer, CTO	Japan	Overall technology roadmap and BiCS strategy.
Masashi Yokotsuka	Managing Executive Officer, VP SSD Division	Japan	Productization of AI-ready storage and AiSAQ™.
Atsushi Inoue	Executive Officer, VP Memory Division	Japan	Flash memory technology innovation and FMS keynote.
Neville Ichhaporia	SVP & GM, SSD Business Unit	USA	US-based SSD strategy and NVIDIA/OCP collaboration.
Masafumi Takahashi	Senior Fellow, Frontier Tech R&D Institute	Japan	Research into AI data center storage architecture.
Rory Bolt	Senior Fellow & Principal Architect	USA	Primary architect for AiSAQ™, SEF, and FDP.
Gili Buzaglo	Director, Software R&D	Israel	Software development for intelligent storage.

European and Greece-Adjacent Presence

KIOXIA Europe GmbH is headquartered in Düsseldorf, Germany, focusing primarily on sales and marketing.

- **Research Presence:** KIOXIA Israel Ltd. serves as a key overseas R&D site for software

and controller techniques.

- **Regional Collaboration Potential:** No dedicated KIOXIA R&D center exists in Greece or Eastern Europe. However, the region is seeing a surge in AI research. The University of Nicosia (UNIC Athens) has partnered with Purdue University for a "living lab" in the Ellinikon smart city project, focusing on AI and data science. Greece's National Recovery and Resilience Plan "Greece 2.0" is establishing 13 Innovation Centres for digital skills and AI, presenting a prime opportunity for academic-industrial collaboration.

Strategic Gaps and Actionable Opportunities

The analysis reveals four critical opportunities where on-device AI for storage can address KIOXIA's current vulnerabilities.

1. Hardening AiSAQ™ into the SSD Controller Silicon

KIOXIA has successfully proven the software-based vector search model through AiSAQ™. The next logical step is to integrate the Product Quantization (PQ) and graph traversal logic into a dedicated hardware block within the next-generation controller ASIC. This would provide a "Search-on-Drive" feature that is significantly faster than the current software-over-NVMe implementation, targeting the 100 million IOPS performance tier.

2. The "Privacy-in-Box" Personal Storage Tier

There is a major opportunity for a client SSD line specifically branded for privacy-conscious AI PC users. By embedding a 1B-parameter BitNet model within a 4TB BG-series SSD, KIOXIA could offer "Self-Indexing" storage that summarizes and tags personal files locally without any OS-level tracking or cloud uploading. This addresses the top concern of IT leaders regarding data leakage while leveraging KIOXIA's leadership in compact form factor (2230) SSDs.

3. CXL-Based Memory Pooling for RAG Infrastructure

The demand for massive memory expansion is currently unmet by DRAM alone. KIOXIA should accelerate its CXL 3.0 controller development to create "XL-Flash Expanders." These devices would present themselves as high-capacity, tiered memory to the host CPU/GPU, using AI-driven prefetching (as seen in the ExPAND project) to mask flash latency. This would allow KIOXIA to capture a portion of the high-margin HBM market by offering a "Memory Expansion as a Service" product.

4. Establishing a Mediterranean R&D Hub for Edge AI

To tap into the European AI research ecosystem and address the lack of Eastern European presence, KIOXIA should consider a collaboration or satellite office in Athens, Greece. The Ellinikon smart city and the UNIC Evolve AI initiative are specifically designed to bridge the gap between US and European innovation. Partnering with these institutes on "One Health" and AI robotics projects would allow KIOXIA to pilot intelligent storage solutions in a "living lab" environment while gaining access to the European Union's National Recovery and Resilience

funding.

Conclusion of Strategic Assessment

KIOXIA is currently at an inflection point. The company's mastery of NAND physics and bit density provides a temporary shield, but the rise of computational storage competitors who are integrating AI-native processing like the ARM R82 indicates that the industry is moving toward "Intelligent Memory". By transitioning from the current software-led AI approach (AiSAQ™) to a hardware-integrated TinyLM strategy, KIOXIA can maintain its 29% market share and redefine the SSD as the cognitive center of the modern data stack.

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